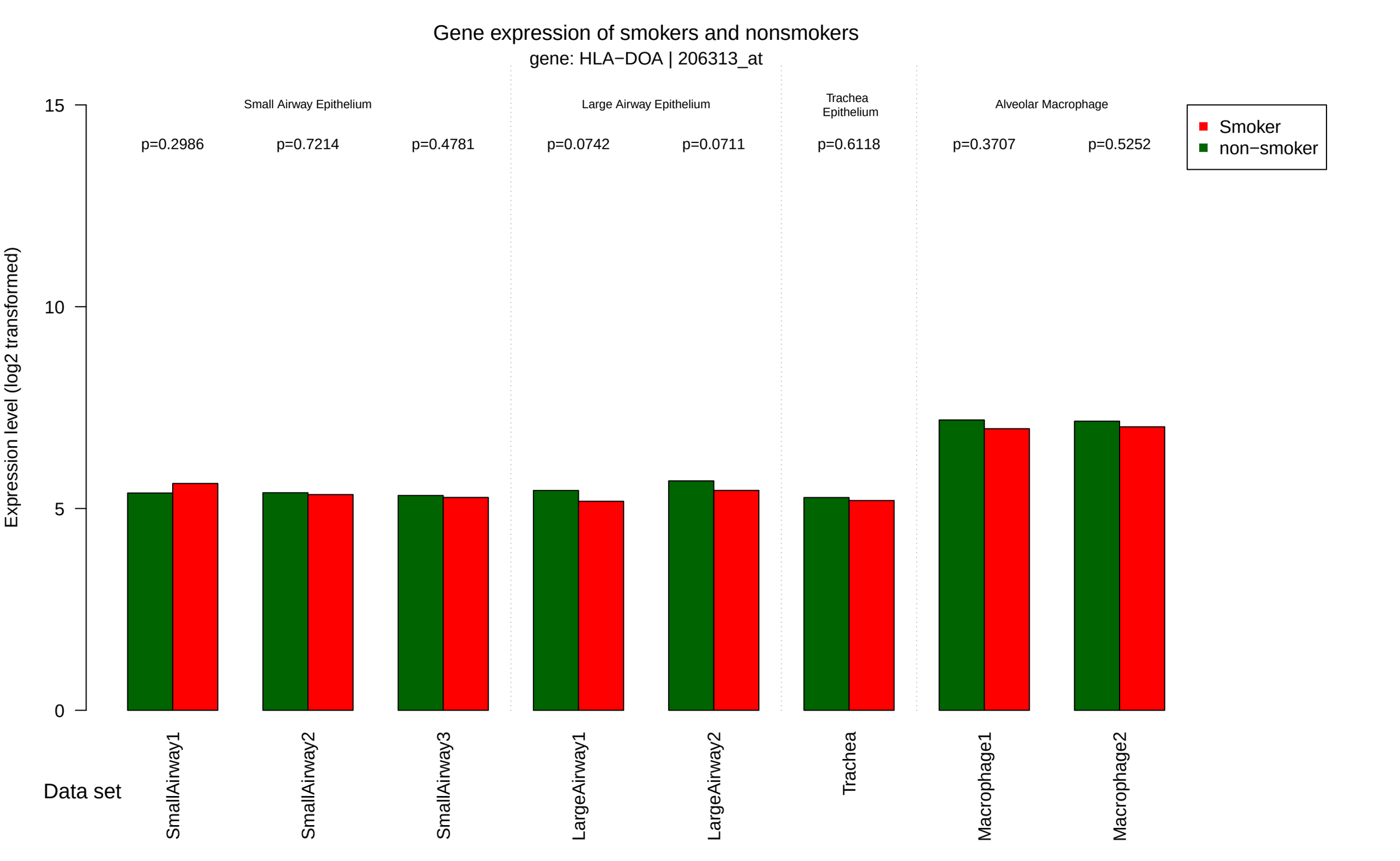
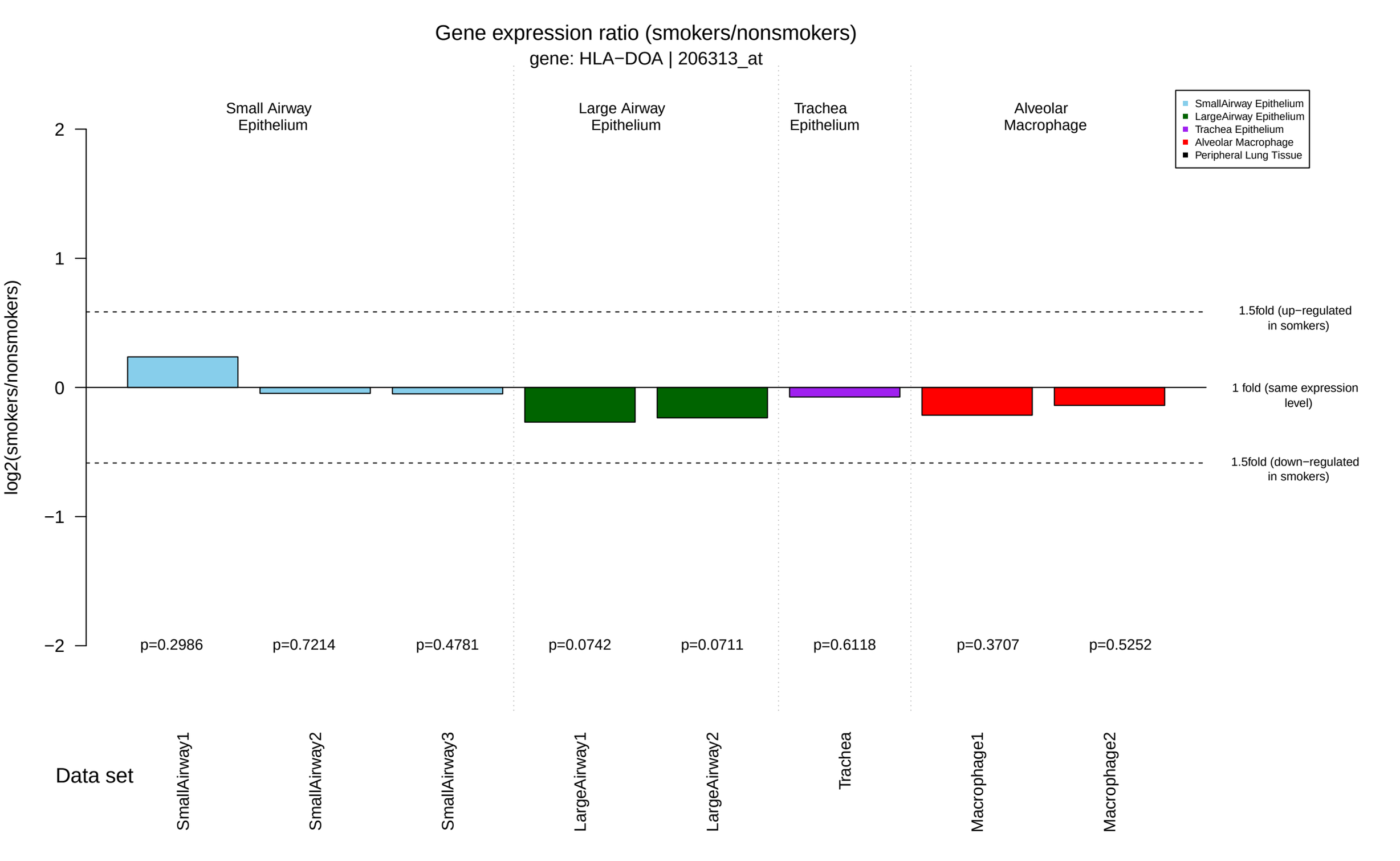
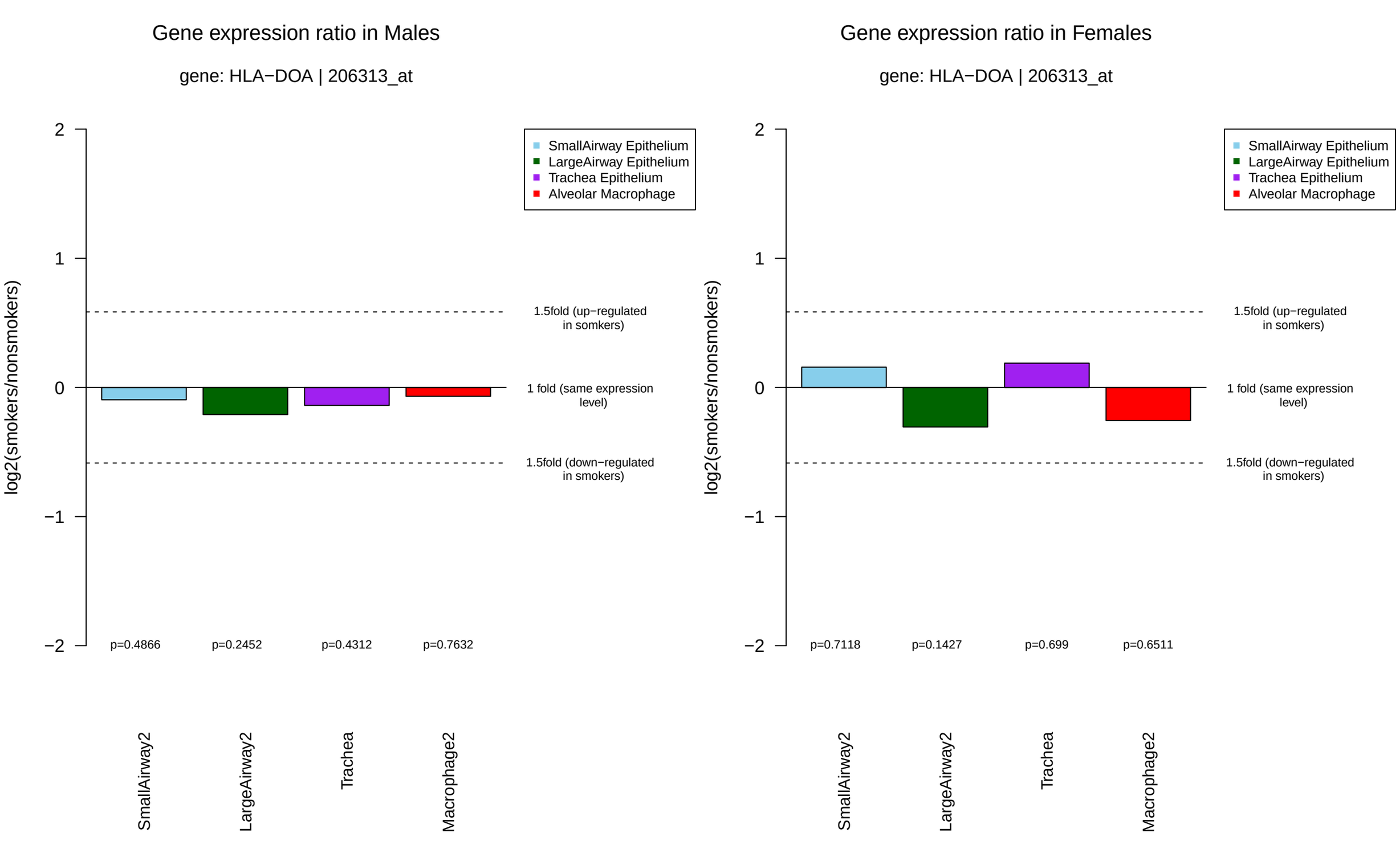
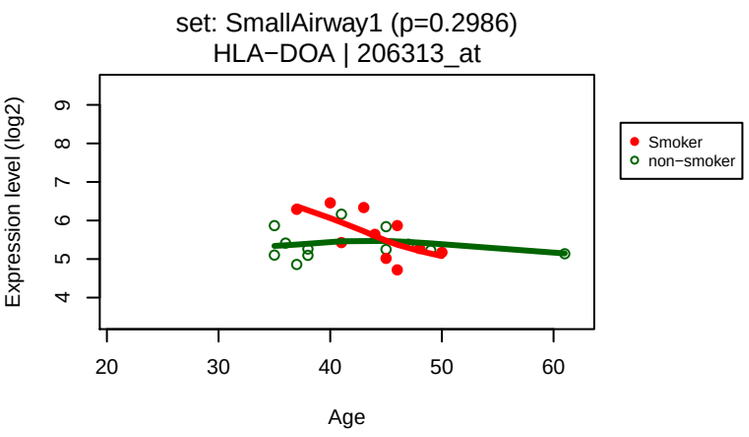
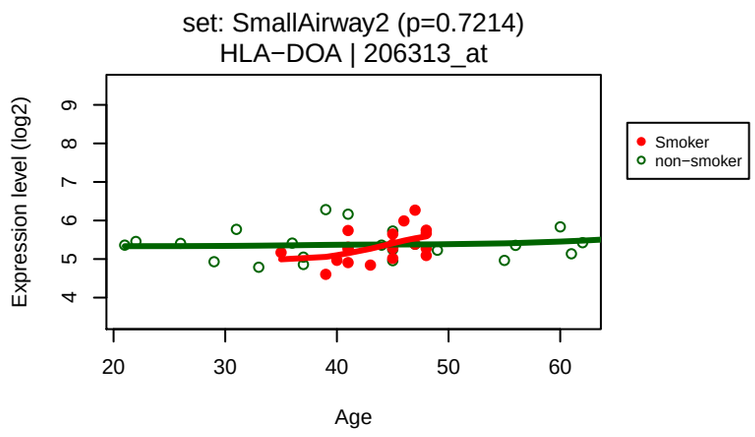
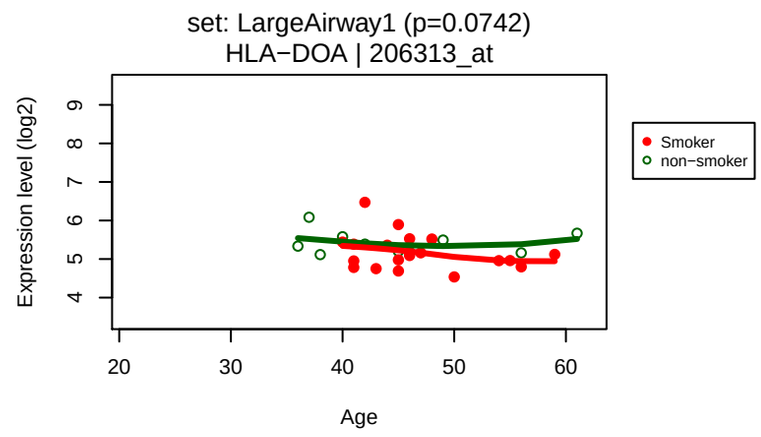

Gene expression of smokers and nonsmokers
gene: HLA−DOA | 206313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2986
p=0.7214
p=0.4781
p=0.0742
p=0.0711
p=0.6118
p=0.3707
p=0.5252
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

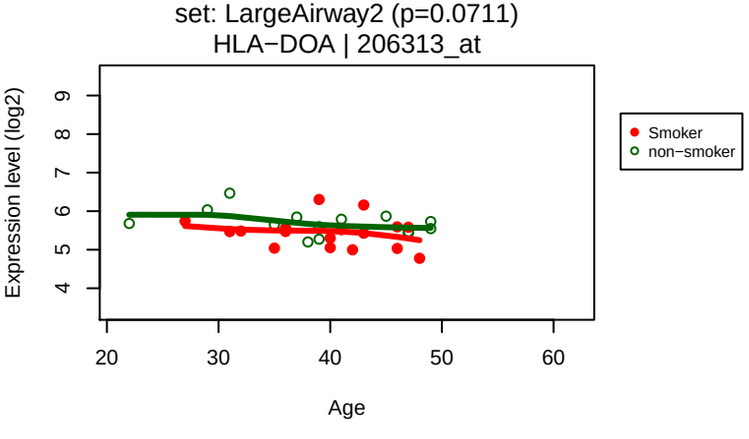

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DOA | 206313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2986
p=0.7214
p=0.4781
p=0.0742
p=0.0711
p=0.6118
p=0.3707
p=0.5252
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

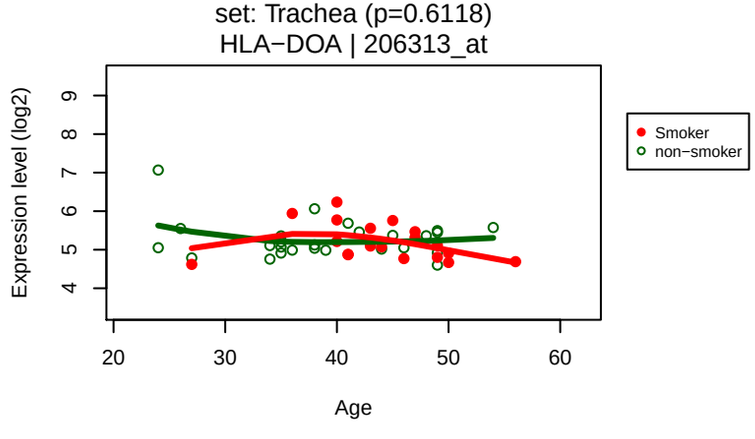

Gene expression ratio in Males
gene: HLA−DOA | 206313_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4866
p=0.2452
p=0.4312
p=0.7632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DOA | 206313_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7118
p=0.1427
p=0.699
p=0.6511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

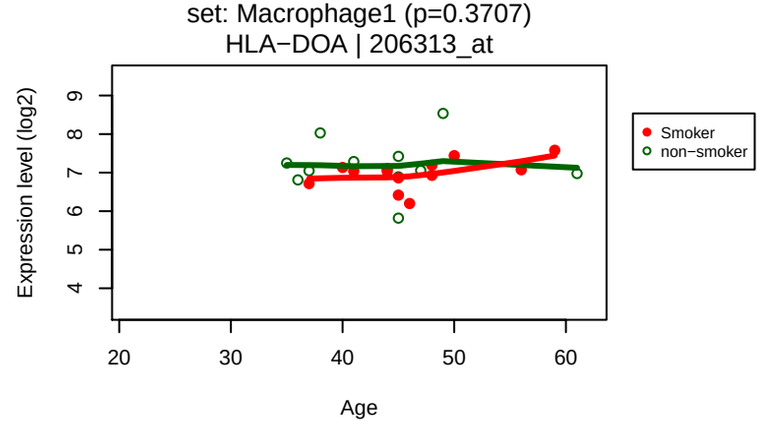

set: SmallAirway1 (p=0.2986)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker

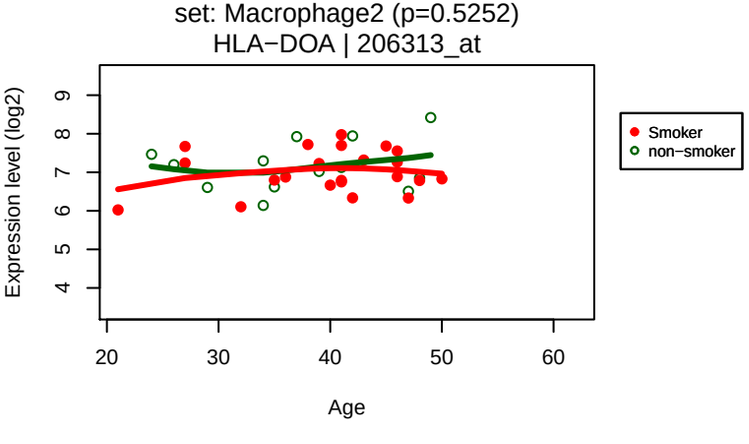

set: SmallAirway2 (p=0.7214)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker

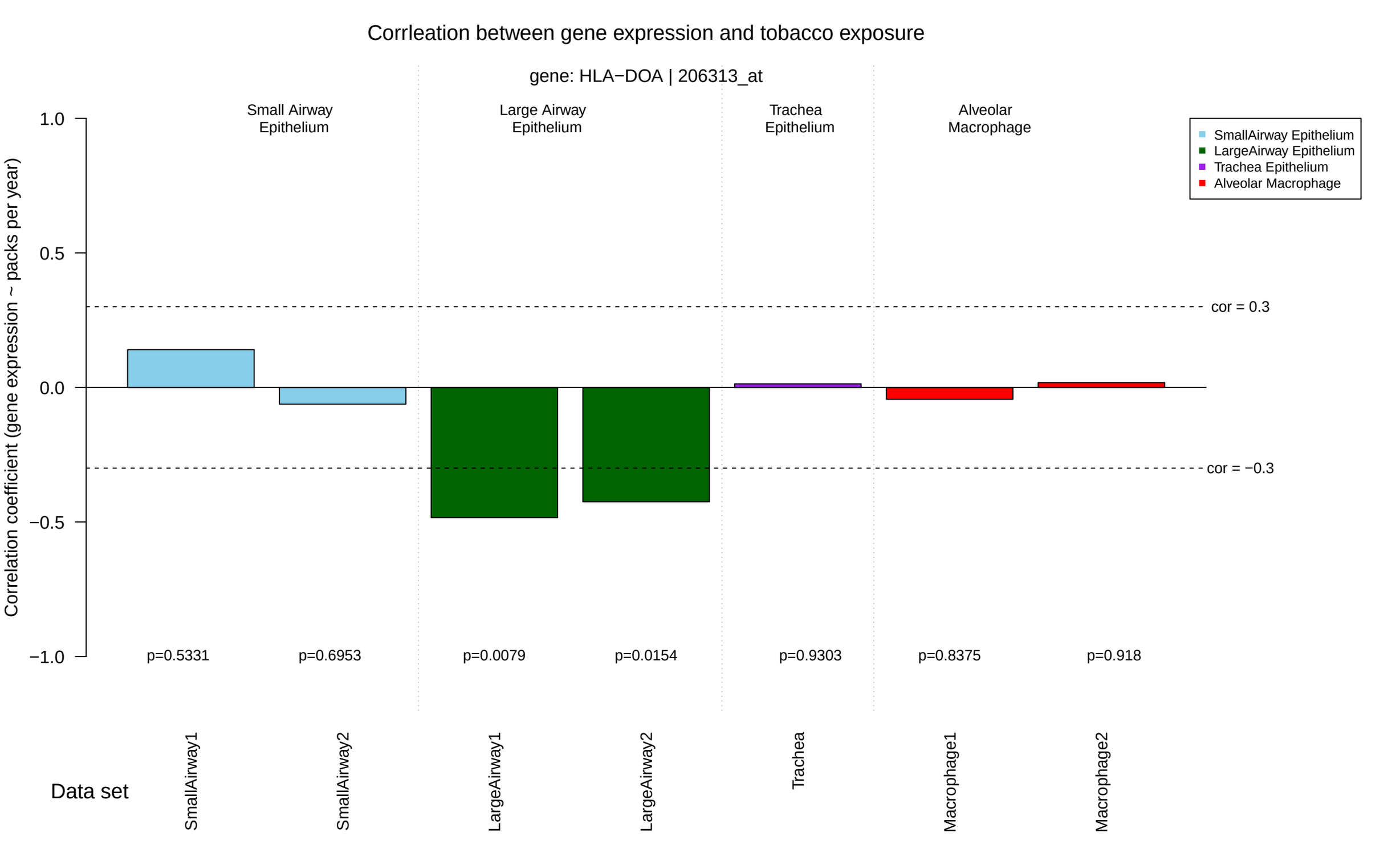

set: LargeAirway1 (p=0.0742)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0711)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6118)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3707)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5252)
HLA−DOA | 206313_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DOA | 206313_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5331
p=0.6953
p=0.0079
p=0.0154
p=0.9303
p=0.8375
p=0.918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2